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HYDROGEN FUELLED GAS TURBINE ENGINE

Abstract

A fuel system for a gas turbine engine configured to combust hydrogen fuel. The fuel system includes a main fuel conduit, a fuel pump configured to operate on hydrogen within the fuel conduit to provide pressurised fuel to a core combustor of the gas turbine engine, an auxiliary combustor downstream in fuel flow of the fuel pump, and configured to combust a portion of fuel diverted from the main fuel conduit and to heat a remainder of fuel in the main fuel conduit, and a fuel turbine downstream in fuel flow of the auxiliary combustor. The fuel turbine is configured to be driven by the heated fuel from the auxiliary combustor and configured to power the fuel pump. The fuel system includes a turbine bypass conduit configured to selectively bypass fuel around the fuel turbine. A method of operation is also described.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates to a fuel system for a hydrogen fuelled gas turbine engine, a gas turbine engine including such a fuel system, aircraft comprising such engines, and methods of controlling such fuel systems and engines.

BACKGROUND

[0002] Hydrogen fuelled aircraft have been proposed, in which the hydrogen is stored in the form of a liquid, in order to improve density of the fuel. However, such hydrogen must first be warmed to higher temperatures and pumped to high pressures at high flow rates prior to combustion.

[0003] Challenges in conditioning the liquid hydrogen include the high power required to raise the temperatures and pressures at the required flow rate, as well as controlling complex pumping and heating systems such that the required flow, pressures and temperatures can be produced at all phases of flight.

SUMMARY

[0004] In a first aspect, there is provided a fuel system for a gas turbine engine configured to combust hydrogen fuel, the fuel system comprising: [0005] a main fuel conduit; [0006] a fuel pump configured to operate on hydrogen within the fuel conduit to provide pressurised fuel to a core combustor of the gas turbine engine; [0007] an auxiliary combustor downstream in fuel flow of the fuel pump, and configured to combust a portion of fuel diverted from the main fuel conduit and to heat a remainder of fuel in the main fuel conduit; and [0008] a fuel turbine downstream in fuel flow of the auxiliary combustor, the fuel turbine being configured to be driven by the heated fuel from the auxiliary combustor and configured to power the fuel pump; wherein [0009] the fuel system comprises a turbine bypass conduit configured to selectively bypass fuel around the fuel turbine.

[0010] Advantageously, the fuel system conditions fuel in an efficient, controllable manner. The provision of a fuel turbine provides for efficient driving power to enable operation of the pump, to raise the pressure of the fuel, while the auxiliary combustor both raises the temperature of the fuel and provides enthalpy to drive the turbine. The provision of the turbine bypass conduit which provides for selective bypass of the fuel turbine provides for independent control of the turbine driven fuel pump, which in turn allows for independent temperature and pump pressure control.

[0011] The fuel turbine may be configured to mechanically drive the fuel pump, and/or may be configured to drive an electrical generator. Advantageously, excess energy from the fuel turbine can be utilised by the aircraft.

[0012] The fuel pump may be coupled to an electric motor. Advantageously, where the fuel turbine cannot provide sufficient power to fully operate the fuel pump, power can be introduced by the electric motor.

[0013] The fuel system may comprise a fuel storage unit configured to store cooled hydrogen, which may be configured to store liquid hydrogen or cryogenically cooled compressed gaseous or supercritical hydrogen.

[0014] The fuel storage unit may be configured to store hydrogen as a liquid at a temperature of less than 30 Kelvin (K), and may be configured to store hydrogen at a temperature less than 25K, and may be configured to store hydrogen at a pressure of between 1 and 4 Bar.

[0015] The fuel pump may be configured to provide a maximum pressure ratio of between 10:1 and 100:1, and may be configured to provide a maximum pressure ratio of between 15:1 and 60:1, and may be configured to provide a maximum pressure ratio of approximately 20:1.

[0016] The fuel system may comprise an actively controllable turbine bypass valve configured to control mass flow rate and/or pressure of fuel flowing through the fuel turbine bypass conduit.

[0017] The fuel system may comprise a turbine inlet valve provided upstream in hydrogen fuel flow of the fuel turbine configured to control mass flow rate and/or pressure of fuel flowing into the fuel turbine. The fuel system may comprise a turbine outlet valve provided downstream in hydrogen fuel flow of the fuel turbine configured to control mass flow rate and/or pressure of fuel flowing out of the fuel turbine. Advantageously, further control of the fuel turbine can be provided, either concurrently or redundantly with the turbine bypass.

[0018] The fuel system may comprise a controller configured to control at least the auxiliary combustor and turbine bypass valve and, optionally, the electric motor, turbine inlet and/or turbine outlet valve to control pressure, temperature and flow rate of fuel within the fuel system.

[0019] The controller may be configured to operate at least one of the auxiliary combustor and turbine bypass valve and, optionally, the electric motor, turbine inlet and/or turbine outlet valve to provide a constant fuel pump rotational speed. Advantageously, as input power by the turbine varies, pump speed is kept constant by additional torque input by the electric motor.

[0020] Alternatively, the controller may be configured to operate at least one of the auxiliary combustor and turbine bypass valve and, optionally, the electric motor, turbine inlet and/or turbine outlet valve to provide a variable pump rotational speed, which varies in accordance with a schedule in accordance with fuel flow demand.

[0021] In a second aspect there is provided a gas turbine engine comprising the fuel system of the first aspect.

[0022] In a third aspect there is provided an aircraft comprising a gas turbine engine of the second aspect.

[0023] In a fourth aspect, there is provided a method of operating a fuel system of the first aspect, the method comprising: [0024] controlling, by a controller of a fuel system of the first aspect, at least auxiliary combustor fuel flow to control auxiliary combustor exit temperature, and a fuel turbine bypass valve position to control fuel turbine speed, such that a required hydrogen pressure, temperature and flow rate is provided downstream of the fuel turbine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] An embodiment will now be described by way of example only with reference to the accompanying drawings, which are purely schematic and not to scale, and in which:

[0026] FIG. **1** shows an airliner comprising a hydrogen powered gas turbine engine;

[0027] FIG. **2** shows a fuel system and gas turbine engine of FIG. **1**;

[0028] FIG. **3** shows a first control method for operating the fuel system of FIG. **2**; and

[0029] FIG. **4** shows a second control method for operating the fuel system of FIG. **2**.

DETAILED DESCRIPTION

[0030] A cryogenic hydrogen fuelled airliner is illustrated in FIG. **1**. In this example, the airliner **101** is of substantially conventional tube-and-wing twinjet configuration with a central fuselage **102** and substantially identical engines **103** which are mounted in an underwing configuration.

[0031] A cryogenic fuel storage system **104** is located in the fuselage **102** and is connected to the engines **103** via a fuel delivery system **105**.

[0032] In the present embodiment, the cryogenic fuel is hydrogen stored as a liquid, below its boiling point. In a specific embodiment, the cryogenic fuel storage system **104** is configured to store the hydrogen fuel at 25 Kelvin. In the present embodiment, the hydrogen fuel is pressurised to a pressure from 1 to 3 bar, and in a specific example, 2 bar. It will be appreciated that the principles of the present invention may be extended to hydrogen fuel stored at higher pressures and/or temperatures and may be applicable to hydrogen stored as either a compressed gas or a supercritical fluid.

[0033] A block diagram of one of the engines **103** is shown in FIG. **2**.

[0034] In the present embodiment, the engine **103** is a turbofan comprising a ducted fan **202** located in a nacelle **204**. The fan **202** receives intake air A and generates two airflows: a bypass flow B which passes axially through a bypass duct **203** and a core flow C which enters the gas turbine core **201**.

[0035] The gas turbine core **201** comprises, in axial flow series, a low-pressure compressor **206**, a high-pressure compressor **208**, a combustor **210**, a high-pressure turbine **212**, and a low-pressure turbine **214**.

[0036] In operation, the core flow C is compressed by the low-pressure compressor **206** and is then directed into the high-pressure compressor **208** where further compression takes place. The compressed air exhausted from the high-pressure compressor **208** is directed into the combustor **210** where it is mixed with fuel and the mixture is combusted.

[0037] Following combustion, the resultant hot combustion products are discharged from the combustor **210** and expand through, and thereby drive, the high-pressure turbine **212** and in turn the low-pressure turbine **214**.

[0038] The fan **202** is driven by the low-pressure turbine **214** via a reduction gearbox **216**. In the present embodiment, the reduction gearbox **216** takes the form of an epicyclic gearbox. In alternative arrangement, the gearbox may be omitted and the engine **103** configured as a direct-drive engine, either in a two-spool or three-spool arrangement.

[0039] Turning now to the hydrogen fuel flow, hydrogen fuel from the hydrogen storage system **104** is provided to a high-pressure fuel pump **224** via a hydrogen fuel conduit **226**, shown by the dashed line in FIG. **2**. Fuel upstream of the high-pressure pump **224** is generally in a liquid state. One or more separate low-pressure pumps (not shown) may also be provided upstream of the high-pressure pump **224**. Though the hydrogen fuel pump **224** is shown as a single component in FIG. **2**, it will be appreciated that the hydrogen fuel pump may comprise a plurality of stages, each configured to raise the pressure of the fuel for delivery to a subsequent stage.

[0040] In general, the fuel pump **224** comprises a multi-stage centrifugal pump configured to provide a fuel pressure ratio of between 10:1 and 100:1 at aircraft maximum take-off power conditions. In other words, the ratio between fuel pressure and atmospheric pressure during take-off is between 10:1 and 100:1. As will be appreciated, maximum take-off thrust for a given engine can be defined as the maximum thrust for which an engine is certified. Typically, this will be achieved at a speed above static, such as approximately Mach 0.25. Maximum thrust may be flat-rated, or may vary according to temperature, with the highest thrust available at any temperature for which the engine is certified defining the maximum take-off thrust. In one embodiment modelled by the inventors, the fuel pump is configured to provide a pressure ratio of approximately 100:1, and provides an absolute pressure of approximately 80 to 120 Bar, and provides fuel at an outlet at a temperature of approximately 33 K or higher. At these conditions, the hydrogen is in a supercritical state.

[0041] Such a high-pressure ratio is required to provide the fuel at the conditions necessary for downstream components, as will be described in more detail below.

[0042] Downstream in hydrogen fuel flow of the pump **224** is a fuel preheater **228**. The fuel pre-heater **228** is configured to raise the temperature of the hydrogen fuel, which is typically still in a liquid phase, downstream of the pump **224**. The pre-heater serves **228** two main roles-firstly, to raise the temperature of fuel downstream to a temperature suitable for combustion in the gas turbine engine core combustor **236**, and also to provide additional enthalpy in the hydrogen fuel flow downstream, for purposes explained below.

[0043] The pre-heater **228** comprises an auxiliary combustor which is configured to draw on a portion of fuel tapped off from the hydrogen fuel conduit **226** via an offtake **229**, and combust this fuel with compressed air drawn from the high-pressure compressor **208** via a bleed air line **230**, shown by the dotted line in FIG. **2**. Combustion gases from the auxiliary combustor are used to heat hydrogen in the conduit **226** via a heat exchanger **232**. Consequently, fuel downstream of the heat exchanger **232** is typically vaporised, and is now in either a gaseous or supercritical phase.

[0044] Fuel flow into the pre-heater **228**, and therefore heat output of the pre-heater **228**, is controlled by a fuel flow valve **231**. One or more bleed valves **233** are typically provided to control bleed air flow into the pre-heat **228**, to provide a desired fuel-air ratio.

[0045] Hydrogen downstream of the pre-heater **228** is typically at substantially the same pressure as hydrogen upstream, with only slight pressure losses being encountered. The temperature of the hydrogen downstream of the pre-heater **228** is raised, with typical temperatures being between 250 and 300 K at maximum take-off conditions, and in one example, 288K.

[0046] A fuel turbine **234** is provided downstream in hydrogen fuel flow of the pre-heater **228**. The fuel turbine **234** is configured to extract mechanical power from the heated, gaseous hydrogen in the fuel conduit **226** downstream of the pre-heater **228**.

[0047] The fuel turbine **234** comprises an axial or centrifugal turbine, which may for instance comprise a lightweight, strong material such as aluminium. Aluminium is chosen because of its high tensile strength, and the relatively low temperatures as the inlet conditions. Additionally, aluminium is resistant to hydrogen embrittlement. Suitable alternative materials include stainless steel.

[0048] A fuel turbine bypass conduit **240** is provided, which fluidly couples an upstream in hydrogen fuel flow side of the turbine **234** to a downstream in hydrogen fuel flow side of the turbine **234**, bypassing the turbine impellor, such that a portion of hydrogen within the conduit **224** can bypass the turbine **234**, without flowing therethrough.

[0049] A bypass valve **242** is provided, which provides for selective control of a mass flow rate of hydrogen flowing through the bypass conduit **240**, and thereby also controls mass flow of hydrogen fuel flow through the turbine **234**. Any suitable valve may be utilised. Examples include throttle valves such as butterfly valves.

[0050] One or more injectors (not shown) are provided downstream in hydrogen fuel flow of the fuel turbine **234**. The injectors provide hydrogen to the combustor **236** at an injection temperature and pressure. Typically, in order to provide sufficient flow into the combustor **210**, the injectors are configured to be operated at an injection pressure ratio of approximately 2:1, and in this example, 2.1:1. The injectors also require fuel at a minimum injector delivery temperature. The minimum injector delivery temperature is defined by a minimum temperature necessary for the hydrogen to provide stable combustion within the combustor **210** in the volume provided. Additional considerations for the minimum delivery temperature include icing. As such, a minimum delivery temperature may comprise 273 K to avoid icing, or may comprise a lower temperature such as 150K to ensure combustion stability.

[0051] The fuel turbine **234** and fuel pump **224** are coupled by a drive arrangement comprising a shaft **238**. The shaft **238** couples the turbine **234** and pump **224** such that they rotate together. An electrical motor **244** is also optionally coupled to the shaft **238**. As such, the fuel turbine **234** at least partly drives the fuel pump **224**, with additional power optionally being provided by the motor **244**. Further loads may also be driven by the fuel turbine **234**, such as an electrical generator **241**.

[0052] Optionally, a turbine outlet throttle valve **248** may be provided, downstream in hydrogen fuel flow of the turbine **234**, and upstream of the combustor **236**. The throttle valve **248** is configured to control mass flow and/or pressure of the fuel flowing therethrough and may comprise any suitable valve such as a butterfly valve. Similarly, an optional fuel turbine inlet throttle valve **247** may be provided upstream in hydrogen fuel flow of the fuel turbine **234**. The throttle valve **248** is configured to control mass flow and/or pressure of the fuel flowing therethrough, and again may comprise any suitable valve such as a butterfly valve. As such, flow rate and pressure both

upstream and downstream of the turbine **234** can be controlled independently or in conjunction with the bypass valve **242**.
[0053] In operation, the engine operates as follows.
[0054] Hydrogen is delivered from the fuel tank **104** to the fuel pump **224**, where fuel pressure is increased. Where the hydrogen is provided from the tank as a liquid, the pump typically maintains the hydrogen in liquid phase. This fuel is delivered to the pre-heater **228**. When the preheater **228** is operational, a portion of the fuel is diverted to the auxiliary combustor **228**, where it is combusted with air, while the remainder is heated by heat exchange with preheater exhaust gases in the preheater. The portion diverted to the auxiliary combustor **228** is controlled by the valve **231**. Where the preheater is non-operational, hydrogen passes through the preheater **228** without being significantly heated. Hydrogen fuel is then delivered to the fuel turbine **234** where it is expanded, and provided to the injectors **206**. A portion of hydrogen fuel may bypass the turbine **234** via the bypass conduit **240**, in accordance with operation of the valve **242**.
[0055] In driving the turbine **234**, torque is generated, which is provided to the pump **224** via the shaft **238**. Additional torque can be provided to the pump **234** from the motor **244** where required.
[0056] Operation of the fuel system is controlled by a controller **250**, which controls at least the valves **231**, **242** to control auxiliary combustor **228** fuel flow and turbine bypass flow respectively. The controller **250** controls at least the valves **231**, **242** and, optionally, the motor **244**, and turbine inlet and outlet valves **247**, **248** such that each of a pressure, flow rate and delivery temperature requirements are met during all stages of flight.
[0057] As will be understood, the minimum pressure requirement must be met such that the compressor delivery pressure P30 can be overcome to permit fuel to flow into the combustor **236**, i.e. there must be a positive pressure gradient between the fuel in the fuel conduit **226** and compressor air delivered to the combustor **236**. On the other hand, an excessively high pressure represents wasted energy in the fuel system and may also result in damage to the fuel system, or excessive flame length, which may in turn damage the turbine **212**, **214**. Similarly, the minimum temperature must be met to avoid icing, and/or provide stable combustion. Again, an excessive fuel temperature represents wasted energy, and may also damage engine components. Finally, the fuel mass flow rate must be controlled to within a relatively small margin (i.e. neither substantially greater than, nor less than a desired value), to provide accurate thrust control of the engine **201**. The rate of change of fuel flow is also important, to provide both acceptable engine acceleration, and to avoid surge and stall.
[0058] FIG. **3** illustrates a method of controlling the fuel system to meet the flow requirements.
[0059] In a first step, the valve **231** is controlled to control fuel flow to the auxiliary combustor **228** to provide target fuel conditions downstream. The hydrogen flows through the turbine **234**, thereby driving the pump **224**. Separate pressure, temperature and flow sensors **252**, **254**, **256** are provided downstream of the fuel turbine **234**, upstream of the injector **236** to determine hydrogen fuel flow conditions at the injector **236**.
[0060] As mentioned, each of a minimum fuel temperature, a minimum combustor entry pressure, and a desired fuel flow rate must be provided.
[0061] In general, the auxiliary combustor **228** is controlled to ensure that the fuel temperature requirements are met, i.e. that the fuel is above a predetermined minimum temperature. This may be controlled on the basis of “closed loop control”, and could utilise conventional control methods, such as Proportional, Integral, Derivative (PID) control.
[0062] In order to control fuel flow rate to a predetermined level, the bypass valve **242**, turbine inlet valve **247** and turbine outlet valve **248** are controlled in accordance with a control schedule.
[0063] In general, the turbine outlet valve **248** is operated as a Fuel Metering Unit (FMU), which is operated in accordance with a schedule based on a commanded fuel flow demand rate. The fuel flow demand rate is in turn determined by a throttle position or auto-throttle demand.
[0064] Where an increased fuel flow demand is made, the turbine outlet valve **248** is opened to

allow increased flow, while the valve **248** is closed to provide reduced flow.
[0065] In order for the outlet valve **248** to have sufficient bandwidth to provide throttle control over the full range of operation, it may be desirable for the pump **224** to operate at a constant speed. As such, a constant pump speed setpoint is defined, and the bypass valve **242** and electric motor **244** are controlled to provide the desired pump speed.
[0066] As such, the turbine bypass **242** can be operated to turn down the work available to the turbine **234**, and thereby the pump **224** speed, with increased bypass flow generally resulting in reduced turbine work, and so reduced pump speed, compared to where the valve **242** is closed, and all hydrogen flows through the fuel turbine **234**. As such, opening the valve **234** will reduce pump **224** work, and so reduce both mass flow and pressure, while closing the valve will increase pump work, and increase mass flow and pressure. Meanwhile, the overall temperature of the fuel downstream of the fuel turbine is generally unaffected, whether the fuel flows through the bypass valve **242**, or the turbine **234**.
[0067] In some cases however, the required pump speed may still not be provided even in the event where the bypass valve **242** is fully closed, and the fuel turbine **234** is operating at full capacity. In such a case, the motor **244** may be operated, to provide additional torque to the pump **224**. In one embodiment, the motor **244** is controlled to provide a variable torque to maintain a constant pump rotational speed setpoint. As such, pressure and flow rate conditions downstream of the pump are less variable than where the motor is not present, and control is greatly simplified. Such a mode of operation may be necessary during start-up, where pressure and flow requirements are relatively low, but little compressor air is available to drive the preheater **228**, and as such, relatively little enthalpy is available to drive the fuel turbine **234**.
[0068] In one case, the turbine inlet valve **247** may also be operated in a similar manner to the bypass valve **242**, to control turbine work. In general, closing the inlet valve **247** will reduce turbine work, and so reduce pump speed. This control could be in addition to operation of the bypass, with the valves **242**, **247** working in concert to control turbine work. Alternatively, the valves **242**, **247** could provide redundant control, ensuring that turbine work control, and therefor pump speed control is assured in the event of the failure of one of the valves **242**, **247**.
[0069] Operating the pump at a fixed speed will guarantee that sufficient pressure head is available for full operation of the valve **248**. However, as will be appreciated, the pressure head will be greater than that required during most operational conditions, leading to wasted pump work. In general, fuel flow demand in large gas turbine engines varies slowly, over the course of several seconds, and so it may be desirable to operate the pump at a variable flow rate. On the other hand, excess work generated by the pump **224** may result in additional heat, which may allow for reduced operation of the pre-heater **228**, and so a low overall drop in efficiency.
[0070] FIG. **4** illustrates an alternative control method. This control method has the advantage that fuel pressure upstream of the valve **248** can be varied.
[0071] In this control method, the valve **248** is operated as before in accordance with fuel flow demand. The valve **248** will operate relatively slowly in accordance with a fuelling schedule, as rapid increase or reduction in fuel flow may result in flameout or turbine damage, as will be understood in the field of gas turbine engine control.
[0072] However, in some cases, the pump **224** can be controlled more rapidly in response to changes in fuel flow demand, in view of its smaller rotational mass. As such, in response to an increase in fuel flow demand, the pump target rotational speed (and so pressure and flow rate) is increased in advance of increases in valve **248** position. Similarly, pump target rotational speed can be reduced in advance of decreases in valve **248** position.
[0073] As shown in FIG. **4** therefore, the system operates in accordance with a modified schedule. The first and second steps are the same as before, as the pre-heater **228** is controlled to maintain fuel temperature.
[0074] Fuel flow demand signals are sent to the controller **250**, which calculates a valve **248**

setpoint position to provide the required flow rate. Before, or as the valve **248** is moved toward the new setpoint, a new pump rotational speed setpoint is also calculated, which will provide sufficient pressure head to the valve **248** to enable a required fuel flow at the valve **248** setpoint. The controller then controls both the valve **248** to control the fuel flow demand, and the valves **247**, **242** and motor **244** to provide the required pump rotational speed.
[0075] As such, less pressure is wasted by over-operation of the pump **224**.
[0076] In one embodiment, the valve **242** may normally be operated in a part open state. As such, the valve can be opened or closed to provide either rapid increase or decrease in fuel flow demand as required.
[0077] Further control is provided by operation of either or both of the throttle valves **247**, **248** and motor **244**. Operation of the throttle valve **247** can provide even more rapid control of fuel flow, at the expense of energy wasted by throttling the flow. As such. Similarly, where insufficient mass flow or pressure is provided where the valve **242** is fully closed, additional pump torque can be provided by the motor **244**.
[0078] Alternative embodiments can be envisaged. For example, the fuel turbine may comprise multiple stages coupled to respective stages of the fuel pump in a similar manner to the multiple compressor and turbine stages of the gas turbine engine. The hydrogen fuel could be stored as a low temperature gas or supercritical fluid. Other control schemes could be envisaged, such as model-based control.

Claims

1. A fuel system for a gas turbine engine configured to combust hydrogen fuel, the fuel system comprising: a main fuel conduit; a fuel pump configured to operate on hydrogen within the fuel conduit to provide pressurised fuel to a core combustor of the gas turbine engine; an auxiliary combustor downstream in fuel flow of the fuel pump, and configured to combust a portion of fuel diverted from the main fuel conduit and to heat a remainder of fuel in the main fuel conduit; and a fuel turbine downstream in fuel flow of the auxiliary combustor, the fuel turbine being configured to be driven by the heated fuel from the auxiliary combustor and configured to power the fuel pump; wherein the fuel system comprises a turbine bypass conduit configured to selectively bypass fuel around the fuel turbine.
2. A fuel system according to claim 1, wherein the fuel turbine is configured to mechanically drive the fuel pump, and/or may be configured to drive an electrical generator.
3. A fuel system according to claim 1, wherein fuel pump is coupled to an electric motor.
4. A fuel system according to claim 1 comprising a fuel storage unit configured to store cooled hydrogen, which may be configured to store liquid hydrogen or cryogenically cooled compressed gaseous or supercritical hydrogen.
5. A fuel system according to claim 4, wherein the fuel storage unit is configured to store hydrogen as a liquid at a temperature of less than 30 Kelvin (K), and may be configured to store hydrogen at a temperature less than 25K, and may be configured to store hydrogen at a pressure of between 1 and 4 Bar.
6. A fuel system according to claim 1, wherein the fuel pump is configured to provide a maximum pressure ratio of between 10:1 and 100:1, and may be configured to provide a maximum pressure ratio of between 15:1 and 60:1, and may be configured to provide a maximum pressure ratio of approximately 20:1.
7. A fuel system according to claim 1 comprising an actively controllable turbine bypass valve configured to control mass flow rate and/or pressure of fuel flowing through the fuel turbine bypass conduit.
8. A fuel system according to claim 1 comprising a turbine inlet valve provided upstream in hydrogen fuel flow of the fuel turbine configured to control mass flow rate and/or pressure of fuel

flowing into the fuel turbine.

9. A fuel system according to claim 1 comprising a turbine outlet valve provided downstream in hydrogen fuel flow of the fuel turbine configured to control mass flow rate and/or pressure of fuel flowing out of the fuel turbine.

10. A fuel system according to claim 7 comprising a controller configured to control at least the auxiliary combustor and turbine bypass valve and, optionally, the electric motor, turbine inlet and/or turbine outlet valve to control pressure, temperature and flow rate of fuel within the fuel system.

11. A fuel system according to claim 10, wherein the controller is configured to operate at least one of the auxiliary combustor and turbine bypass valve and, optionally, the electric motor, turbine inlet and/or turbine outlet valve to provide a constant fuel pump rotational speed.

12. A fuel system according to claim 10, wherein the controller is configured to operate at least one of the auxiliary combustor and turbine bypass valve and, optionally, the electric motor, turbine inlet and/or turbine outlet valve to provide a variable fuel pump rotational speed, which varies in accordance with a schedule in accordance with fuel flow demand.

13. A gas turbine engine comprising a fuel system according to claim 1.

14. An aircraft comprising a gas turbine engine according to claim 13.

15. A method comprising controlling, by a controller of a fuel system in accordance with claim 1, at least auxiliary combustor fuel flow to control auxiliary combustor exit temperature, and a fuel turbine bypass valve position to control fuel turbine speed, such that a required hydrogen pressure, temperature and flow rate is provided downstream of the fuel turbine.